

Fundamentals of Generative AI and Large Language Models (LLMs)

A Technical Overview of Concepts, Architectures, and Real-World Applications

Author: Gayathri Murugan

Classification: Academic Technical Report

Table of Contents

1. Executive Summary
2. Introduction to Generative AI
3. Core Fundamentals of LLMs
4. Architecture of Large Language Models
5. Training and Scaling
6. Applications of Generative AI
7. Challenges and Limitations
8. Future Scope and Innovations
9. Conclusion
10. References

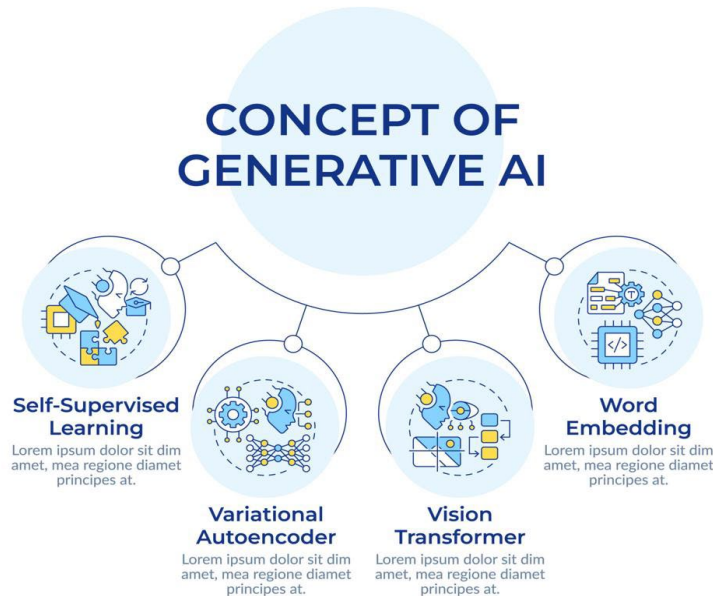
1. Executive Summary

This report presents the **fundamentals of Generative Artificial Intelligence and Large Language Models (LLMs)**. It explains how modern AI systems generate human-like text, code, and multimedia content using advanced neural network architectures. The report covers tokenization, embeddings, transformers, attention mechanisms, training methods, and practical applications. It also highlights current challenges such as hallucinations, bias, and computational costs.

2. Introduction to Generative AI

Generative AI is a branch of artificial intelligence focused on **creating new content** rather than only analyzing existing data. It can generate:

- Text



- Images
- Audio
- Video
- Program code

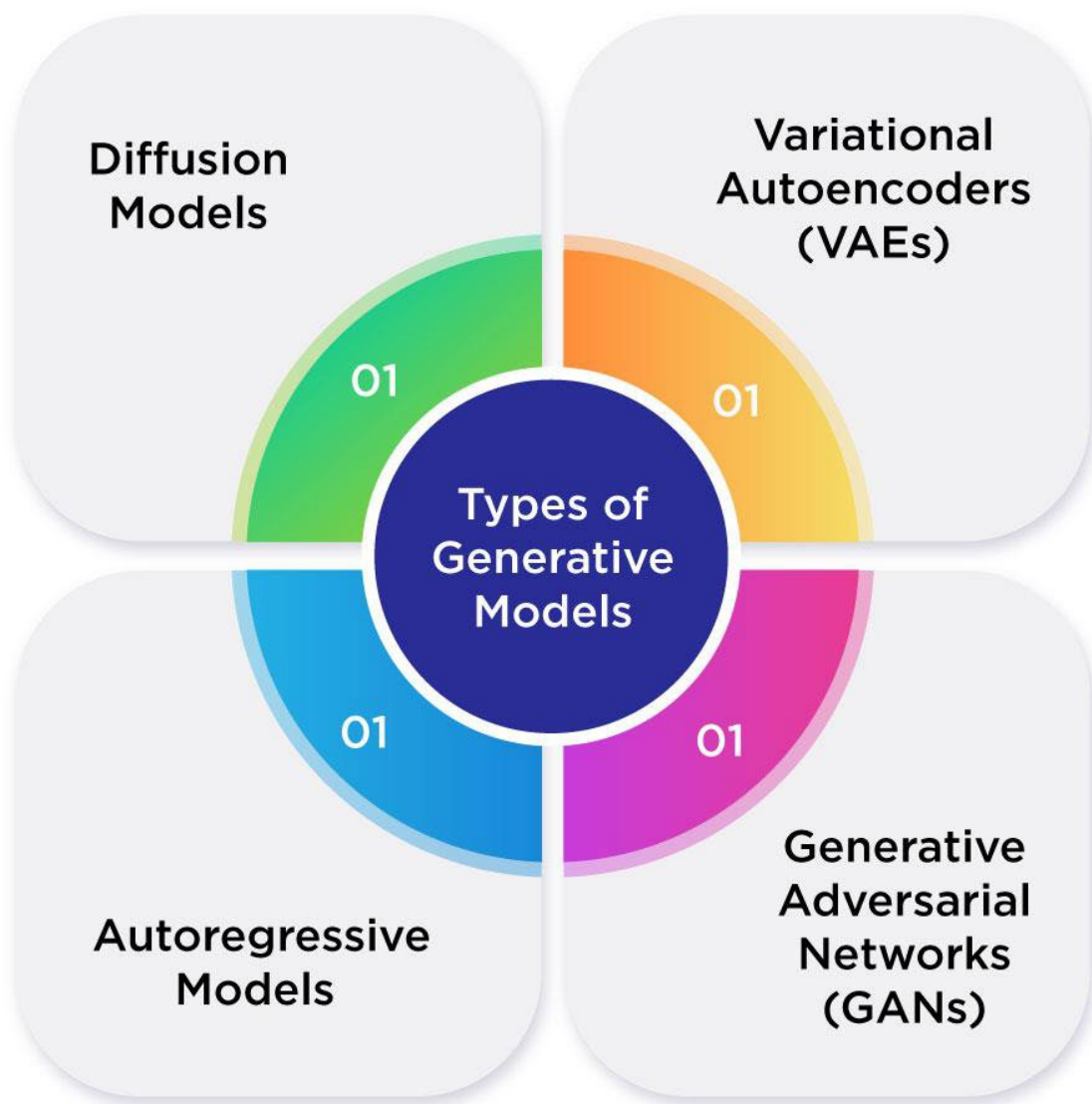
Unlike traditional AI models that classify or predict labels, generative systems produce entirely new outputs based on learned patterns.

2.1 Traditional AI vs Generative AI

Feature	Traditional AI	Generative AI
Purpose	Classification	Content Creation
Output	Labels / Scores	Text, Images, Code
Learning	Supervised	Self-supervised

Feature	Traditional AI	Generative AI
Example	Spam Filter	ChatGPT

3. Core Fundamentals of LLMs



3.1 What is an LLM?

A Large Language Model is an AI model trained on massive datasets to understand and generate human language.

3.2 Tokens

Text is divided into smaller units called **tokens**. Example: > “Artificial Intelligence” → [Artificial] [Intelligence]

3.3 Embeddings

Tokens are converted into numerical vectors called **embeddings**, which help the model understand meaning and relationships between words.

3.4 Probability Prediction

LLMs predict the **next most likely token** in a sequence.

Example: > “The sun rises in the ____”

The model predicts **east** based on probability.

4. Architecture of Large Language Models

4.1 Transformer Model

Most modern LLMs are based on the **Transformer architecture** introduced in *Attention Is All You Need*.

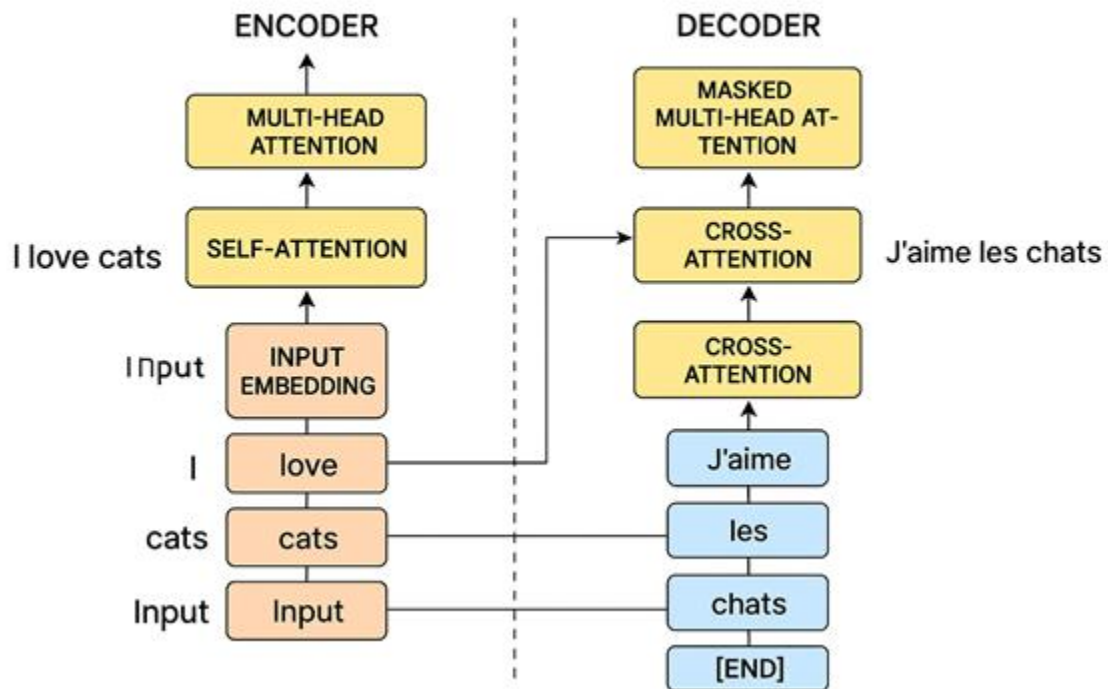
4.2 Self-Attention

Self-attention helps the model understand how words in a sentence relate to each other.

For example: > “Gayathri submitted her report because she completed it early.”

The model understands that **she** refers to **Gayathri**.

Transformer Architecture



4.3 Attention Formula

Attention mechanism is mathematically represented as:

$$[\text{Attention}(Q,K,V)=\text{softmax}()V]$$

5. Training and Scaling

5.1 Pre-training

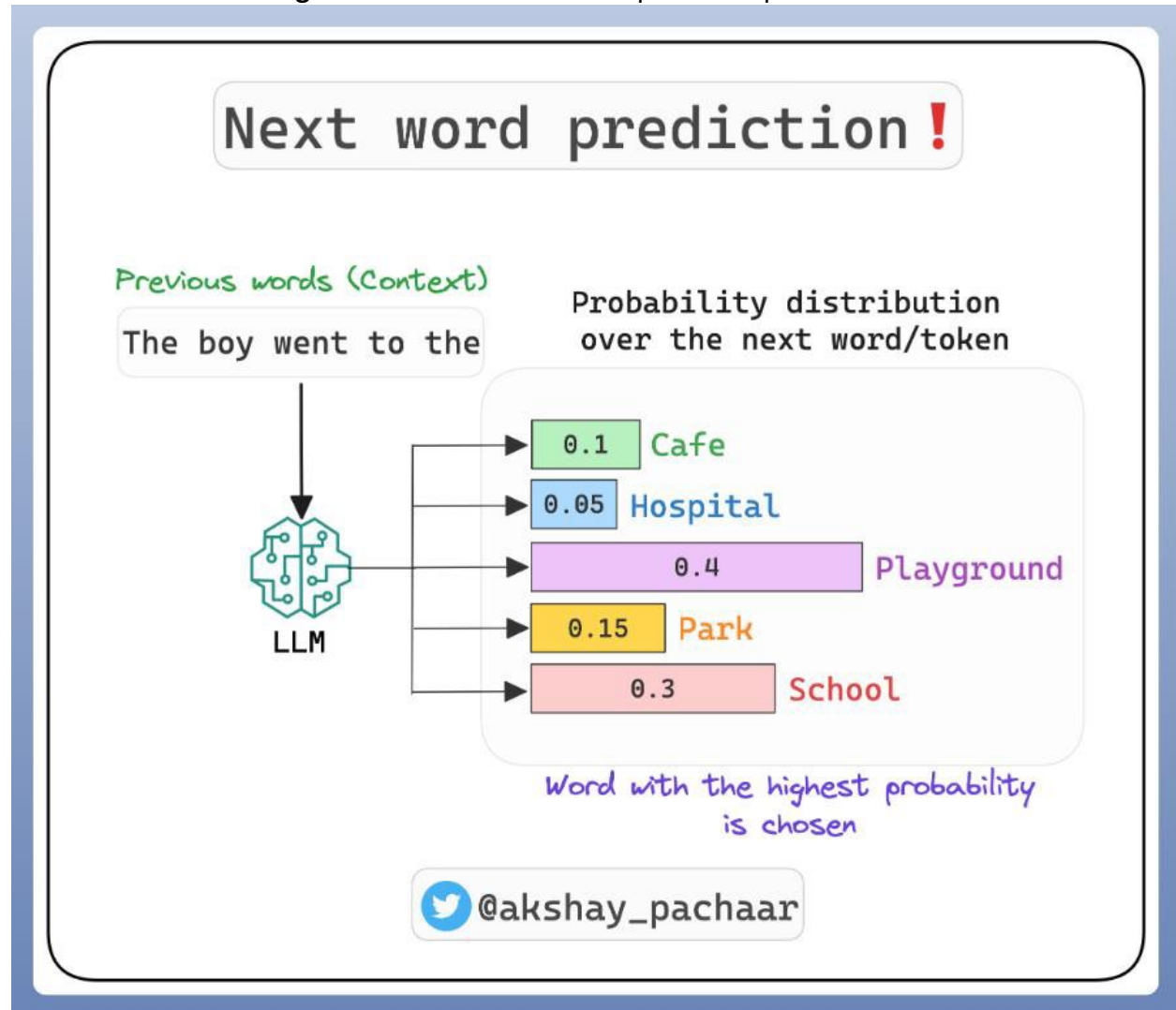
The model learns language patterns from huge datasets such as books, websites, and code repositories.

5.2 Fine-Tuning

The model is further trained for domain-specific tasks.

5.3 RLHF

Reinforcement Learning from Human Feedback improves helpfulness a



nd safety.

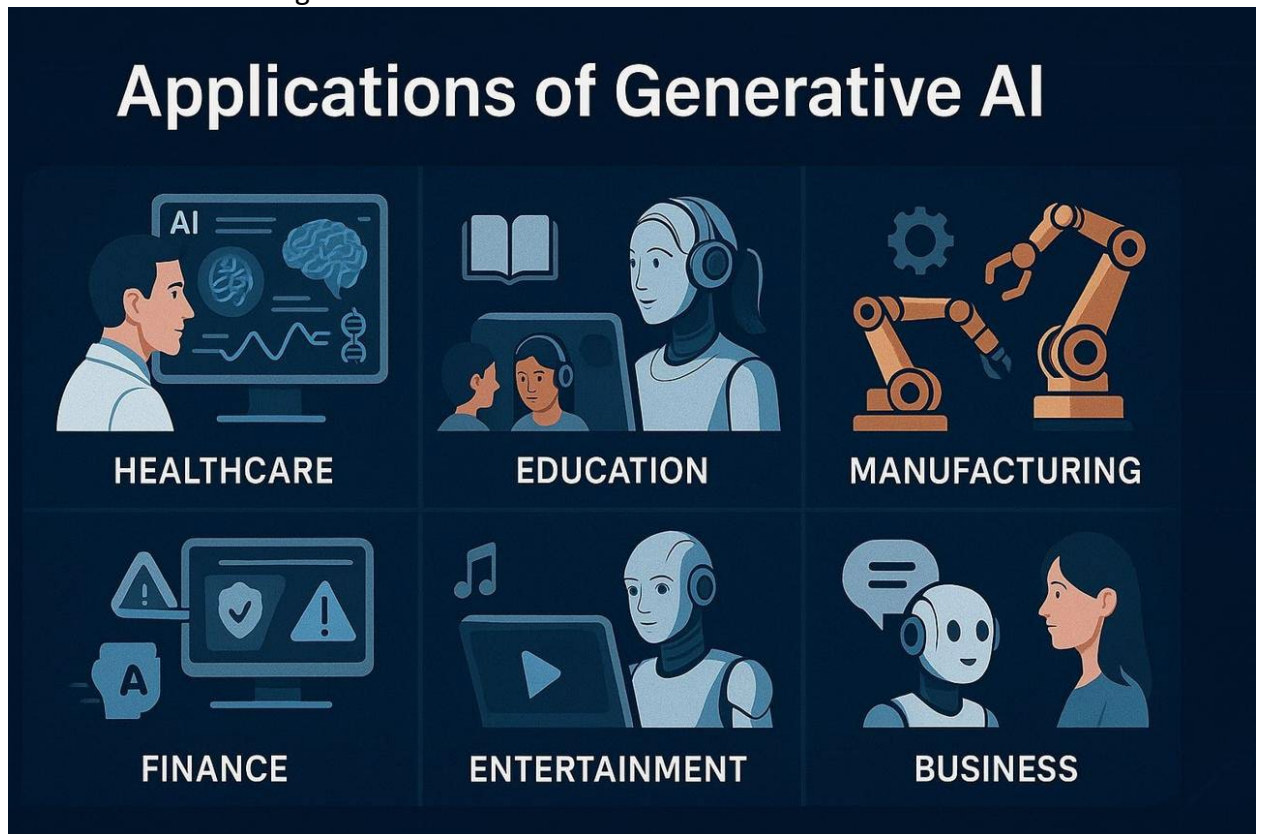
5.4 Scaling

Performance improves with:

- More parameters
- More data
- More compute power

6. Applications of Generative AI

- **Education** – AI tutoring and content summarization



- **Healthcare** – medical report assistance
- **Software Development** – code generation and debugging
- **Marketing** – content writing and personalization
- **Design** – image and video generation

7. Challenges and Limitations

7.1 Hallucination

The model may generate factually incorrect information.

7.2 Bias

Training data may contain societal biases.

7.3 High Computational Cost

Training LLMs requires powerful GPUs and significant energy.

8. Future Scope and Innovations

Future developments include:

- Multimodal AI
 - AI Agents
 - Personalized assistants
 - More efficient smaller models
-

9. Conclusion

Generative AI and LLMs are transforming the way humans interact with technology. Their ability to understand and generate natural language makes them powerful tools across industries.

10. References

1. Vaswani et al. (2017) – Attention Is All You Need
2. Brown et al. (2020) – Language Models are Few-Shot Learners
3. OpenAI Research Papers
4. DeepMind Scaling Law Publications